ScamShield AI Fraud and Scam Awareness Assistant

Summative assessment documentation and Dialogflow test evidence

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CRS Name: Artificial Intelligence	**Course Name:** Artificial Intelligence: Real-World Applications and Implications
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Project Repository: https://github.com/aarav343/scam-shield-ai-sa	**Dialogflow ES agent (Google access required):** https://dialogflow.cloud.google.com/#/editAgent/scamshieldai-k9we/

1 Executive summary and problem

ScamShield AI helps people respond safely to suspicious payment requests, phishing links, impersonation, fake jobs, and related fraud. It gives short warnings, asks for the action requested by the sender, and points people toward an official bank or reporting route when needed. It is an advisory chatbot: it does not verify whether a message is genuine or collect real account credentials.

The live Dialogflow ES agent contains 21 intents and 10 custom entities. Thirteen recorded English scenario checks passed, and the later English greeting screenshot matched Welcome. Hindi is selectable in the agent; the recorded Hindi scam requests reached fallback or an unrelated intent. A separate Hindi package was not active during those observations.

2 Intended users and design decisions

The intended users include digital payment customers, students and job seekers, older adults, and people who may already have sent money or shared a credential. The design focuses on urgency cues, unfamiliar links, advance fees, and requests for OTPs or UPI PINs. These user groups informed the prototype design.

The assistant asks users to describe warning signs without pasting OTPs, PINs, passwords, CVV, or full account details. It gives immediate caution before requesting more detail, and it uses follow-up context for scam details and the safety quiz.

3 Dialogflow implementation

The English agent uses 19 custom intents plus the default Welcome and Fallback intents. Each custom intent has 12 example phrases in the prepared import. The 10 custom entities standardise terms such as payment method, contact channel, requested action, and loss status. The prototype uses built-in responses and short-lived Dialogflow contexts; it does not need a webhook or external data store for these responses.

Custom entity catalogue

Entity	Example values
@scam-type	UPI fraud, phishing, job scam, investment scam, loan scam, impersonation
@contact-channel	WhatsApp, SMS, email, phone call, social media, website

Entity	Example values
@requested-action	share OTP, send money, open link, install app, scan QR, share PIN
@credential-type	OTP, UPI PIN, password, CVV, card number
@payment-method	UPI, QR code, bank transfer, card, wallet
@impersonated-identity	bank, employer, courier, support
@urgency-cue	immediately, today, within ten minutes, account will close
@link-feature	shortened link, misspelled domain, unknown website, unexpected attachment
@job-fee	registration fee, training fee, security deposit, equipment fee
@loss-status	money sent, credentials shared, app installed, no action taken

Intent and entity mapping

Intent	Action or purpose	Custom entities
Default Welcome Intent	Greeting and entry point	—
ScamCheckIntent	scam.check	@contact-channel, @scam-type, @urgency-cue
ScamDetailsIntent	scam.details	@requested-action
OTPIntent	scam.otp	@credential-type, @impersonated-identity, @urgency-cue
UPICollectIntent	scam.upi_collect	@payment-method, @credential-type
QRCodeIntent	scam.qr	@payment-method, @contact-channel
PhishingLinkIntent	scam.phishing	@link-feature, @contact-channel, @urgency-cue
FakeJobIntent	scam.job	@job-fee, @contact-channel, @urgency-cue
ImpersonationIntent	scam.impersonation	@impersonated-identity, @contact-channel, @urgency-cue
InvestmentIntent	scam.investment	@scam-type, @contact-channel, @urgency-cue
LoanAppIntent	scam.loan	@scam-type, @contact-channel, @credential-type
RemoteAccessIntent	scam.remote_access	@requested-action, @impersonated-identity, @loss-status
AccountCompromisedIntent	scam.compromised	@loss-status, @credential-type, @payment-method
ReportFraudIntent	scam.report	@scam-type, @payment-method, @loss-status
PreserveEvidenceIntent	scam.evidence	@contact-channel, @link-feature
SafePaymentTipsIntent	scam.tips	@payment-method, @scam-type

Intent	Action or purpose	Custom entities
QuizStartIntent	quiz.start	—
QuizSafeAnswerIntent	quiz.correct	—
QuizUnsafeAnswerIntent	quiz.incorrect	—
GoodbyeIntent	input.goodbye	—
Default Fallback Intent	Safe recovery from an unmatched input	—

4 Training examples and responses

The table gives one representative training phrase and a concise description of the configured English response for each custom intent. The import package contains the full set of example phrases and exact response text.

Intent	Sample phrase	Configured response excerpt
ScamCheckIntent	Can you check if this message is a scam?	I cannot verify a message as safe from a description alone. Do not click links, pay, or share codes. Tell me what the sender asked you to do, without pasting any OTP, PIN, password, or full account…
ScamDetailsIntent	They asked me to share my OTP	That request is a warning sign. Pause contact and independently check the organisation through its official app or website. Do not share a PIN, OTP, password, or CVV. If you already lost money, contact…
OTPIntent	Should I share my OTP with someone?	Never share an OTP, UPI PIN, password, or CVV with another person, even if they claim to be from your bank. Stop the conversation and contact your bank using a number from its official app or card.
UPICollectIntent	They sent me a UPI collect request	Be careful with unexpected UPI collect requests. Entering your UPI PIN authorises a payment; it is not needed to receive money. Decline the request and verify the sender through an official channel.
QRCodeIntent	They sent a QR code for my refund	Do not scan an unknown QR code or enter your UPI PIN to receive a refund or prize. Scanning and approving with a PIN can make a payment. Verify through the official app instead.
PhishingLinkIntent	I received a shortened link in SMS	Do not open an unexpected link or attachment. A shortened URL, misspelled domain, or urgent login demand is a warning sign. Visit the organisation's official website by typing the address yourself.
FakeJobIntent	A job offer asks for a registration fee	A job offer asking for an upfront registration, training, or equipment fee is a warning sign. Do not pay or share sensitive documents. Verify the employer on its official website and contact the…
ImpersonationIntent	Someone claiming to be my bank called me	A name, logo, or caller ID does not prove identity. End the conversation and contact the bank, government office, employer, courier, or support

Intent	Sample phrase	Configured response excerpt
		team through an independently found official channel. Do…
InvestmentIntent	This investment promises guaranteed returns	Guaranteed high returns and pressure to invest quickly are warning signs. Do not send money based on a chat group or influencer claim. Independently verify the provider and its regulator details before…
LoanAppIntent	A loan app wants an advance fee	Be cautious with loan apps that demand an advance fee, unnecessary phone permissions, or payment outside the official app. Verify the lender independently and do not share an OTP or install unknown…
RemoteAccessIntent	Support asked me to install an app	Do not install a remote-control app for an unsolicited caller. It may let them view your screen or operate your device. If you already installed one, disconnect the device, remove the app, contact your…
AccountCompromisedIntent	I shared my OTP already	Act quickly: contact your bank or payment provider using its official app, website, or card number. Block affected access, change passwords from a trusted device, and review transactions. For financial…
ReportFraudIntent	How do I report online fraud?	If money was lost in India, contact your bank immediately and call the national financial cyber fraud helpline 1930. You can also report at https://cybercrime.gov.in/. Use official sites only. Keep…
PreserveEvidenceIntent	What evidence should I save?	Save screenshots of the message, sender details, links, transaction IDs, dates, and times. Do not delete the chat before reporting. Do not paste private credentials into this chatbot. Share evidence…
SafePaymentTipsIntent	Give me safe payment tips	Before paying, verify the recipient independently, check the amount in your payment app, and avoid links from strangers. Never enter a UPI PIN to receive money and never share OTPs, PINs, passwords, or…
QuizStartIntent	Start a safety quiz	Quick quiz: a caller claims to be your bank and asks for your OTP to stop fraud. Should you share it? Reply yes or no.
QuizSafeAnswerIntent	No	Correct. Never share an OTP with a caller, even if they claim to be from your bank. For help, end the call and use the bank's official contact details.
QuizUnsafeAnswerIntent	Yes	The safe answer is no. A caller asking for your OTP is a warning sign. Never share it; end the call and contact your bank through an official number.
GoodbyeIntent	Goodbye	Stay safe. If you have lost money or shared credentials, contact your bank promptly and report financial cyber fraud through official channels.

5 Live test results in Dialogflow

The English results below were observed in Dialogflow ES Try it now. A pass means the intended intent matched and the observed advice was safety aligned for that input; it does not prove every possible wording or integration path.

Scenario	Observed match	Result
UPI collect request	UPICollectIntent	Pass
OTP impersonation	OTPIntent	Pass
Fake job fee	FakeJobIntent	Pass
Phishing link	PhishingLinkIntent	Pass
Remote access	RemoteAccessIntent	Pass
Money already lost	AccountCompromisedIntent	Pass
Scam detail slot filling	ScamCheckIntent → ScamDetailsIntent	Pass
Quiz, correct answer	QuizStartIntent → QuizSafeAnswerIntent	Pass
Quiz, unsafe answer	QuizStartIntent → QuizUnsafeAnswerIntent	Pass
Fallback	Default Fallback Intent	Pass
UPI refund PIN request	UPICollectIntent	Pass
Misspelled suspicious SMS link	PhishingLinkIntent	Pass
Quiz answer out of context	Default Fallback Intent	Pass
English greeting with en selected	Default Welcome Intent	Pass

The latest screenshot shows the input “hi” and the configured English Welcome response when English was selected in Dialogflow.

Hindi checks before importing the bilingual package

Input	Observed match	Observation
नमस्ते	Default Welcome Intent	Generic Hindi greeting
बैंक वाला मेरा ओटीपी मांग रहा है	Default Fallback Intent	Scam warning did not match
मुझे पैसे लेने के लिए यूपीआई पिन डालने को कह रहे हैं	Default Fallback Intent	UPI warning did not match
मुझे एसएमएस में संदिग्ध लिंक मिला है	Default Welcome Intent	Misrouted to greeting
सुरक्षा क्विज़ शुरू करो	Default Fallback Intent	Quiz did not start
मैंने ठग को पैसे भेज दिए	Default Fallback Intent	Loss guidance did not match

6 Evidence from the live agent

The following screenshots were supplied from the current ScamShieldAI Dialogflow ES console. They document the visible entity list, first page of the intent list, one Hindi fallback, and the English Welcome result.

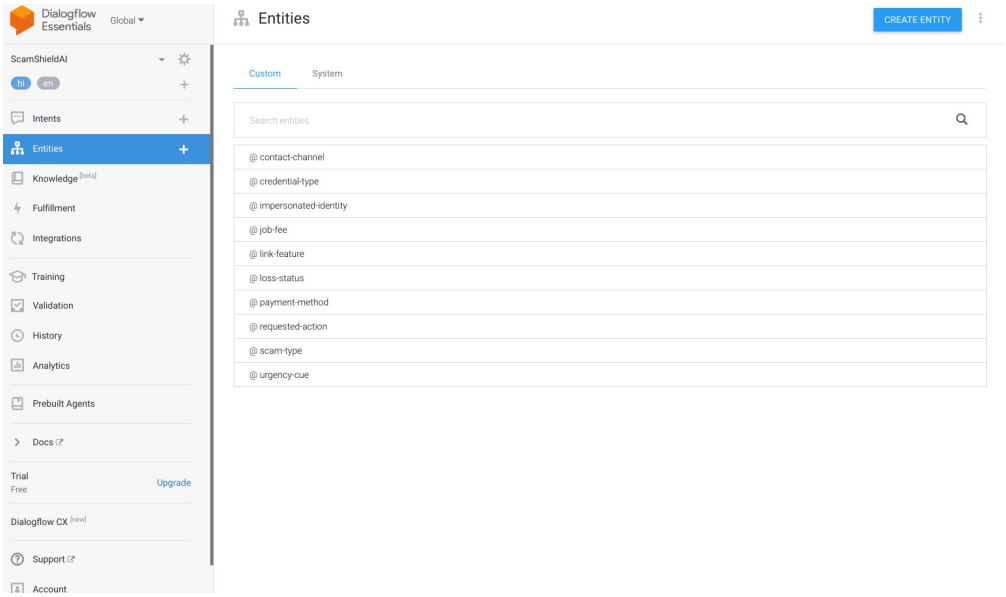


Figure 1 Ten custom entities

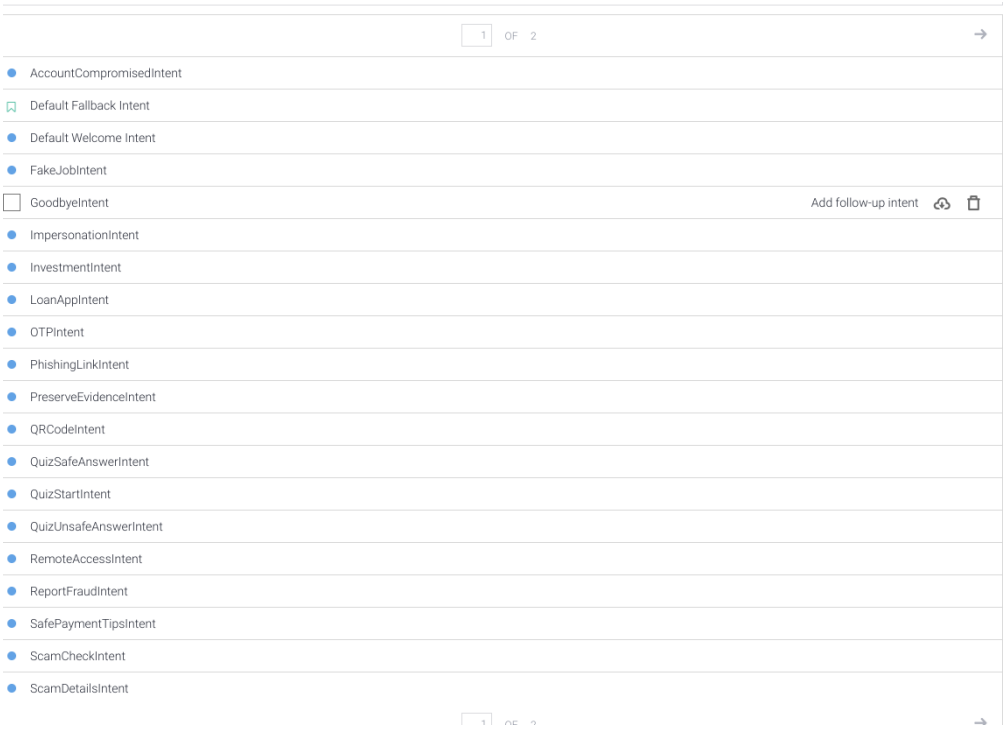


Figure 2 First page of the intent list

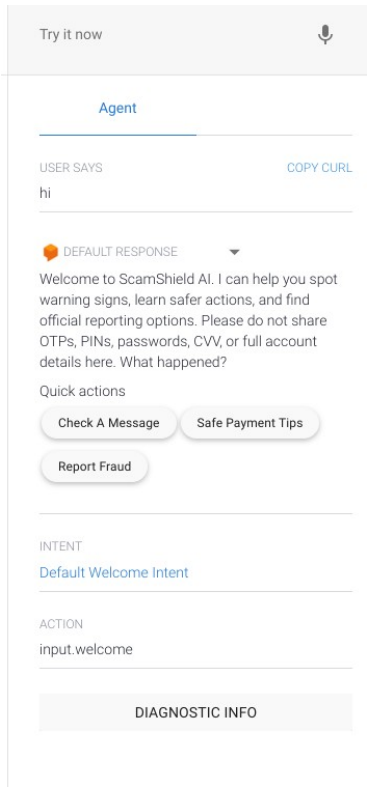


Figure 3 English greeting matched Welcome



Figure 4 Hindi money-loss input reached Fallback